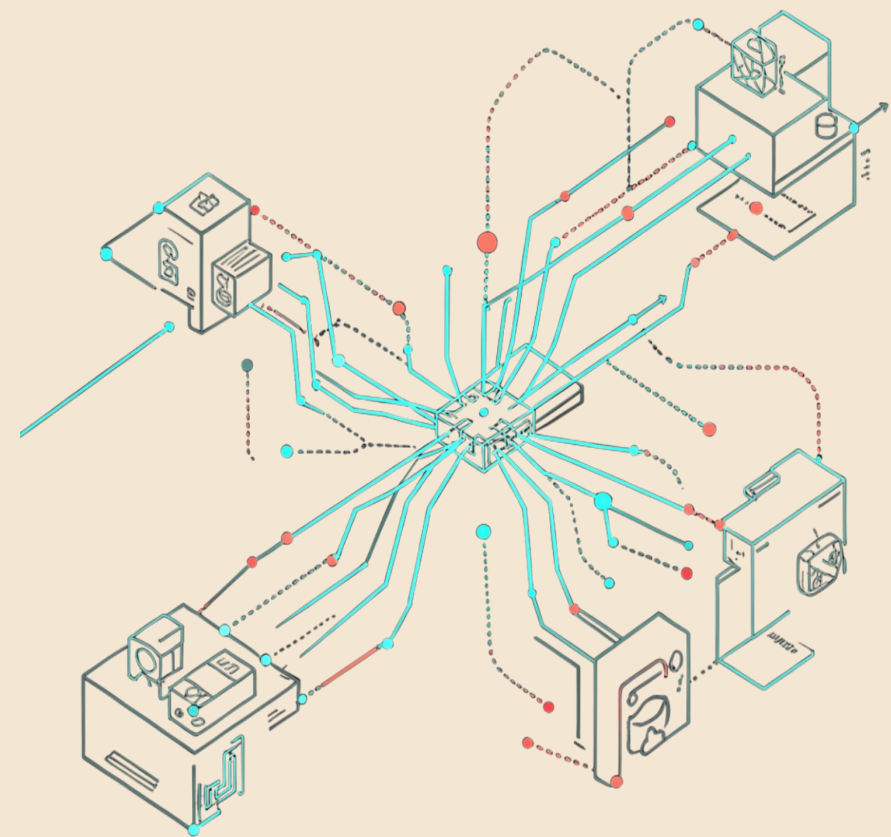
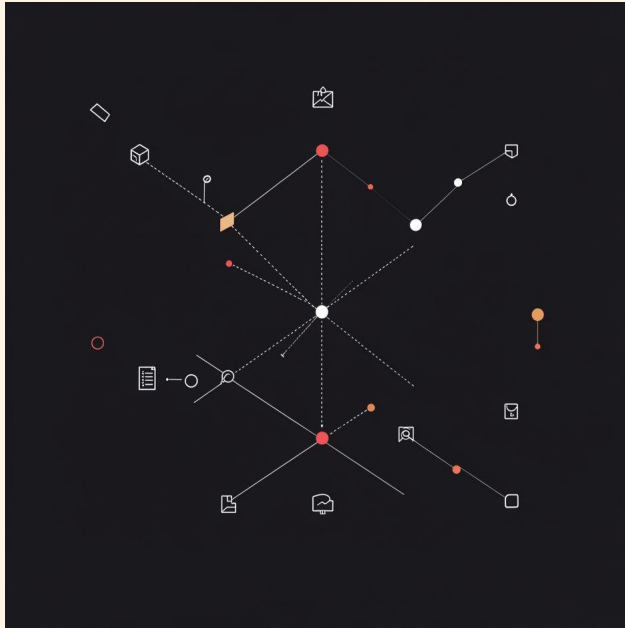




Logistic Regression in Machine Learning

Classification Using Probability-Based Predictive Modeling



What Is Logistic Regression?

→ Definition

A supervised learning algorithm that predicts the **probability** of a categorical outcome — not a continuous value

→ Why "Regression"?

Models a regression equation internally, then maps output to a probability between 0 and 1 via the **Sigmoid function**

→ Binary Classification

Assigns inputs to one of two classes — Yes/No, 0/1, True/False

→ Real-World Examples

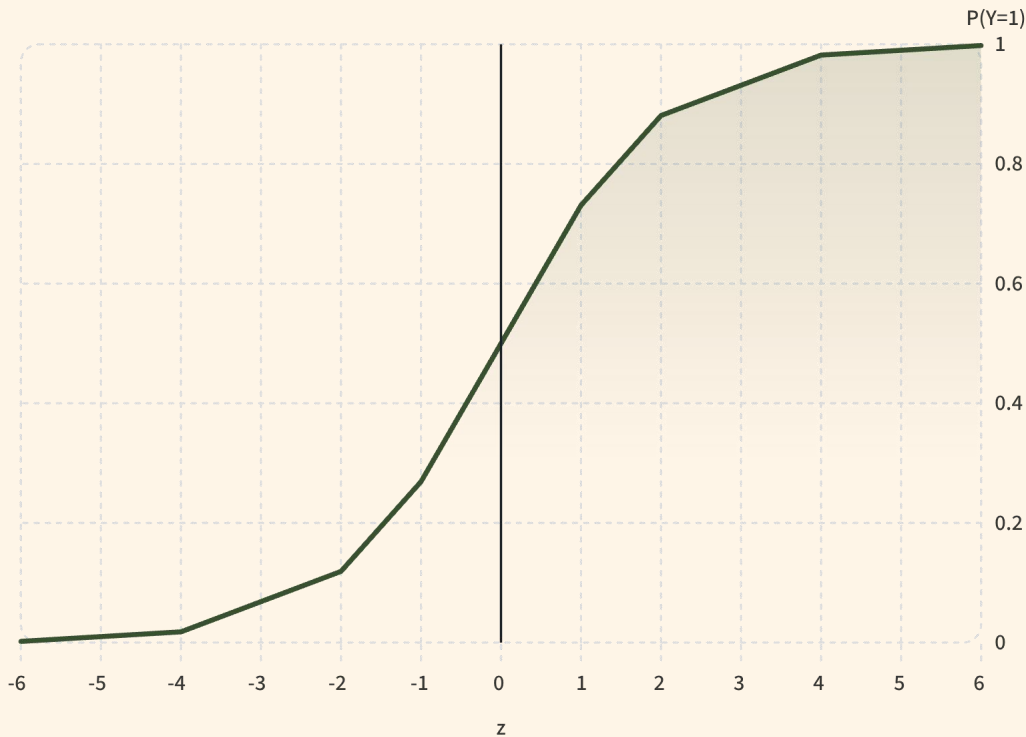
Spam detection · Disease prediction · Loan approval · Customer churn · Email classification

How Logistic Regression Works



Each input feature is weighted and summed into a linear score $\mathbf{z}$, which the Sigmoid function squashes into a probability. A decision threshold of 0.5 converts that probability into a final class label.

The Mathematical Model



Linear Equation

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Sigmoid Function

$P(Y=1) = 1 / (1 + e^{-z})$ — maps any real number to (0, 1)

Coefficients (β)

β_0 = intercept; $\beta_1 \dots \beta_n$ = feature weights learned during training

Decision Boundary

The hyperplane where $P = 0.5$ ($z = 0$) separates the two classes

Types of Logistic Regression

Binary

2 Classes • No Order

Spam / Not Spam, Disease / No Disease

Email filtering, Disease detection

Multinomial

3+ Classes • No Order

Cat / Dog / Horse

Image classification, NLP tasks

Ordinal

3+ Classes • Ordered

Poor → Average → Good → Excellent

Survey ratings, Risk levels

Type	# of Classes	Class Order	Example	Common Application
Binary	2	No	Spam / Not Spam	Email filtering, Disease detection
Multinomial	3+	No	Cat / Dog / Horse	Image classification, NLP
Ordinal	3+	Yes	Poor → Excellent	Survey ratings, Risk levels

Model Training & Evaluation

Training the Model

Uses **Maximum Likelihood Estimation (MLE)** to find optimal β coefficients. Minimizes the **Log-Loss** cost function via **Gradient Descent**.

Accuracy & F1 Score

Accuracy = correct predictions / total. **F1 Score** balances Precision and Recall — critical for imbalanced datasets.

ROC-AUC

ROC curve plots True Positive Rate vs False Positive Rate. **AUC closer to 1.0** = stronger model performance.

Confusion Matrix

	Predicted: Positive	Predicted: Negative
Actual: Positive	TP (True Positive)	FN (False Negative)
Actual: Negative	FP (False Positive)	TN (True Negative)

Assumptions of Logistic Regression

Binary Outcome

Dependent variable must be categorical — binary for standard logistic regression

Independence

Observations must be independent of each other — no repeated measures or clustering

No Multicollinearity

Independent variables should not be highly correlated with one another

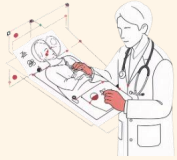
Log-Odds Linearity

Requires a linear relationship between features and the log-odds of the outcome

Large Sample Size

Requires sufficient data to train reliably; sensitive to outliers and small samples

Real-World Applications



Healthcare

Disease diagnosis, cancer detection, patient risk scoring



Finance

Credit scoring, fraud detection, loan approval decisions



Marketing

Customer churn prediction, ad click-through prediction



Cybersecurity

Spam filtering, network intrusion detection



HR

Employee attrition prediction and workforce planning

Advantages, Limitations & Challenges

Aspect	✔ Advantages	⚠ Limitations / Challenges
Complexity	Simple & interpretable	Fails on nonlinear data
Output	Probabilistic (0–1)	Linear decision boundary only
Speed	Fast to train	Needs feature engineering
Data	Works with small datasets	Sensitive to class imbalance & outliers



Simple & Fast

Easy to implement and interpret



Probability Output

Returns calibrated probabilities, not just labels



Linear Only

Cannot capture complex nonlinear boundaries



Class Imbalance

Requires careful handling of skewed datasets

Summary & Key Takeaways

01

Supervised Classification Algorithm

Logistic Regression predicts **probabilities** for categorical outcomes

02

Powered by the Sigmoid Function

Maps linear combinations to the range **(0, 1)** for probability estimation

03

Comprehensive Evaluation

Measured with **Accuracy, Precision, Recall, F1 Score, and ROC-AUC**

04

Universally Applied

Trusted across **healthcare, finance, cybersecurity, and marketing**

"Logistic Regression is one of the most widely used classification algorithms because of its simplicity, interpretability, and strong predictive performance for binary classification problems."

